Sanchita Thayagaran

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EDUCATION

University of Massachusetts Amherst Sep 2024 - May 2026
M.S. Computer Science, *Computer Science*

Anna University, Chennai Aug 2018 - May 2022
B.E. Computer Science, *Computer Science*

SKILLS

Programming Languages: Python, Java, TypeScript, JavaScript, Bash, SQL, C++

Cloud & Distributed Systems: AWS (EC2, Lambda, S3, EKS, RDS, IAM, VPC, CloudWatch), Azure Entra ID SSO, Docker, Kubernetes, Linux (RHEL), Terraform (IaC), GitOps, Secrets Management, Microservices, RESTful APIs, Load Balancing, CI/CD (Jenkins, GitHub Actions)

SRE & Observability: SLI/SLO Design, Incident Response, Root Cause Analysis, Grafana, Prometheus, Splunk, CloudWatch, AppDynamics, OpenTelemetry, Distributed Tracing, PagerDuty, On-Call Operations, Post-Mortem Facilitation, Capacity Planning, Error Budgets, ETL/ELT Pipeline Design, Autosys Job Scheduling, MS SQL Server, Unix, Linux, Windows, JIRA, Confluence

Frameworks & Libraries: Django REST, FastAPI, Spring Boot, Proxy API Architecture, JWT Authentication, RBAC, Microservices Design Patterns, React, Next.js

Databases: PostgreSQL, MySQL, MongoDB, MS SQL Server, Row-Level Locking, SQL-Driven Validation

Security: Azure Entra ID SSO, JWT, RBAC, Least-Privilege IAM, Secrets Management, Secure VPC Networking

AI & Machine Learning: Claude Code, Cursor, Copilot, Agentic Development Workflows, Spec-Driven Development, Prompt Engineering, AI Output Validation for Production, XGBoost, Random Forest, SMOTE, Stratified K-Fold Cross-Validation, Feature Engineering, Python Data Pipelines, SQL Analytics, Apache Kafka, Apache Airflow

Methodologies: Agile, SRE Practices, GitOps, Blameless Post-Mortems, On-Call Rotation, System Design, OOP, Design Patterns

WORK EXPERIENCE

Greenpoint Global | Software Engineer & Developer Fellow | Rye, NY Aug 2026 - Present

Python/JavaScript/Power Bi/Excel

- Lead a 4-member Mapping & Automation team, translating business requirements into technical deliverables, coordinating task ownership and code/review workflows, and driving data-mapping and financial-report automation initiatives with cross-functional stakeholders.
- Engineer configuration-driven data transformation and ETL workflows using Python, SQL, Excel, and structured mapping rules to standardize heterogeneous financial data across loans, deposits, investments, and general-ledger records for downstream reporting systems.
- Migrated and validated client financial records across 6 data domains with zero transformation errors, implementing reusable mapping logic and end-to-end Contract-to-GL and Contract-to-Balance-Sheet reconciliation with traceable validation and exception reporting.
- Develop the Call Report Automation & Mapping Framework for NCUA regulatory reporting, integrating dynamic mapping and standardized data-ingestion workflows to support varying client file structures and reduce reliance on manual, client-specific transformations.

UMass Amherst CSCF | Software Engineer | Amherst, MA Aug 2025 - May 2026

Django/Next.js/React/PostgreSQL

- Independently designed, built, and maintained a full-stack production billing platform for UMass CSCF that processes employee time-based billing for research groups against grant funding, actively used in production by UMass research groups.
- Architected a proxy API security layer between the Next.js/React frontend and Django REST backend, so browser requests are routed through frontend-owned API routes that forward to the backend and relay responses back, keeping backend logic and data never directly exposed to the client.
- Implemented Azure Entra ID SSO via NextAuth, a customized JWT-based authentication and permission mechanism on the backend, and role-based access control across four user roles (Admin, Finance Admin, Grant Admin, PI) enforcing a multi-step bill review and approval workflow.

- Built agentic development workflows using AI coding agents in agent mode, applying spec-driven development and prompt engineering to scaffold the proxy API architecture and multi-role permission logic; validated all AI-generated output for correctness and security before shipping to production.

Morgan Stanley | Software Engineer, Site Reliability & Production Engineering | Bangalore, India Jun 2022 - Aug 2024

Python/Java/Microservices

- Designed, built, and operated Python and Java microservices on AWS (Lambda, EKS, EC2, S3, RDS, IAM) with RESTful APIs, including load balancer configuration and traffic distribution, Docker containerized deployments via Jenkins and GitHub Actions CI/CD, and Terraform-compatible Infrastructure as Code (IaC) automation with GitOps practices and secrets management, while administering Autosys job scheduling and supporting MS SQL Server databases across Unix, Linux, and Windows environments serving Bloomberg and Refinitiv financial data pipelines at scale.
- Built and orchestrated production ETL data pipelines using Apache Airflow, scheduling and monitoring ingestion and transformation workflows for Bloomberg and Refinitiv financial data with automated data quality checks and SQL-driven validation.
- Built observability infrastructure from scratch using Grafana, Prometheus, Splunk, AppDynamics and CloudWatch; defined SLOs and error budgets across owned services; authored 25+ operational runbooks; drove a 30% reduction in mean time to resolution (MTTR) through systematic root cause analysis and blameless post-mortems.
- Eliminated 40+ daily manual operational checks by designing Python and Bash automation pipelines with structured alerting, reducing engineering toil by 45 minutes per engineer per day, while resolving high-severity production incidents under 24/7 on-call rotations across Engineering, Operations, and Platform teams and applying least-privilege IAM policies, secrets management, and secure VPC networking to maintain 99.9%+ uptime across mission-critical distributed systems.

Accolite Digital | Software Development Engineer | Bangalore, India Feb 2022 - Jun 2022

Django/Springboot

- Built RESTful APIs in Python (Django) and Java (Spring Boot) with PostgreSQL and MySQL on AWS; applied design patterns and software engineering best practices.
- Delivered unit-tested Docker containerized microservices via CI/CD pipelines in an Agile development environment.

Tata Consultancy Services Ltd. | Data Science Engineer | Coimbatore, India Jun 2021 - Aug 2021

Python/SQL/ML

- Built an end-to-end Python data pipeline on 3M+ financial records achieving 96.2% AUC and a 22% reduction in false negatives using XGBoost, Random Forest, and SMOTE class balancing, with SQL-driven validation and automated reporting.

PROJECTS

SafeRx - GenAI Medication Safety System

- Built a GenAI medication-safety system using Azure AI Foundry's GPT-4o with a six-step deterministic reasoning chain and hard safety-escalation rules that override model output when a query requires resolution outside the LLM.

Apollo Healthcare Platform

- Designed and built a full-stack distributed application end-to-end: Django REST backend, React frontend, PostgreSQL with row-level locking, AWS Lambda/SNS asynchronous event-driven processing, Google OAuth, and complete CI/CD pipelines from scratch.

5-Stage CPU Pipeline & Cache Simulator

- Built a 5-stage RISC pipeline simulator with hazard detection, branch handling, and cache simulation; identified and resolved a critical pipeline and cache stall bug, achieving a 3.76x speedup over the single-cycle baseline.

Industry 4.0 Healthcare Research Intelligence

- Built an NLP pipeline applying LDA topic modeling and KeyBERT keyword extraction to identify 16 distinct latent topics across four healthcare sub-domains (cardiology, oncology, orthopedics, health records), surfacing technology-adoption trends in AI diagnostics, robotic surgery, and blockchain records management, with topic quality evaluated via coherence scoring.

CERTIFICATIONS & ACHIEVEMENTS

- McKinsey.org Forward Program - McKinsey.org Sept 2026
- Google IT Support - Coursera, Google Sept 2020
- Python for Data Science - NPTEL Aug 2019